

AUTOTRAQ Production Readiness Analysis

Analysis Date: February 1, 2026 **MVP Deadline:** February 3, 2026 **Status:** Functional but incomplete

Executive Summary

AUTOTRAQ is a well-structured auto parts inventory system with solid foundations. The core MVP workflow is functional (parts → fitments → inventory → requests → fulfill), but several routes are incomplete, features are missing when compared to production inventory systems, and there are gaps in both frontend UX and backend robustness.

Current Architecture

Layer	Technology	Status
Frontend	React + Vite + TypeScript	✓ Good
Backend	Express.js + TypeScript	✓ Good
ORM	Prisma	✓ Good
Database	MySQL 8.0	✓ Good
Auth	JWT + bcrypt	✓ Good
Validation	Zod	✓ Good

What Works Well

1. **Append-only inventory ledger** - Properly auditable, source of truth
2. **Role-based access control** - admin, manager, fulfillment, viewer
3. **Request workflow state machine** - PENDING → APPROVED → FULFILLED
4. **SKU generation system** - Structured SKU with barcode generation
5. **Part-vehicle fitment** - Many-to-many with proper cascading
6. **Interchange groups** - Explicit part equivalency tracking
7. **Year >= 2000 validation** - Domain rule enforced
8. **Modern UI** - Clean dark theme, collapsible sidebar

Issues Found

A. Missing/Broken Routes

Route	Issue	Priority
PUT /vehicles/:id	MISSING - Cannot update vehicles	HIGH
DELETE /vehicles/:id	MISSING - Cannot delete vehicles	HIGH
PUT /locations/:id	MISSING - Cannot rename locations	MEDIUM
DELETE /locations/:id	MISSING - Cannot delete locations	MEDIUM
DELETE /interchange-groups/:id	MISSING - Cannot delete entire groups	MEDIUM
POST /inventory/return	MISSING - RETURN event type exists but no endpoint	HIGH
PUT /users/:id	MISSING - Cannot update profile/password	MEDIUM
POST /auth/forgot-password	MISSING - No password reset	LOW
GET /users	MISSING - No user listing for admins	LOW
POST /requests/:id/reject	MISSING - Only approve exists, no reject	MEDIUM

B. Frontend Gaps

Page	Issue	Priority
LoginPage	No register form/toggle - new users can't sign up from UI	HIGH
PartsPage	No pagination - will break with large datasets	MEDIUM
PartDetailPage	Works well, good inline editing	✓ OK
VehiclesPage	Doesn't show which parts fit each vehicle	MEDIUM
InventoryPage	Search nav item exists but not implemented	LOW
RequestsPage	Missing "Cancel" button, only Approve/Fulfill	HIGH
ScanPage	Works well for camera, USB, manual	✓ OK
All pages	Missing loading skeletons, no error boundaries	LOW

C. Features Common in Similar Apps (Missing Here)

Feature	Typical Use	Priority
Low stock alerts	Notify when qty < threshold	HIGH
Supplier management	Track where parts come from	MEDIUM
Purchase orders	Order from suppliers	MEDIUM
Part images	Visual identification	MEDIUM
Location transfers	Move stock between locations	MEDIUM
Export (CSV/PDF)	Reports, data export	MEDIUM
Dashboard/analytics	At-a-glance metrics	LOW
Customer management	Track who requests parts	LOW
Multi-user same request	Collaborative editing	LOW

D. Production Readiness Issues

Issue	Current State	Fix Needed
Logging	console.log only	Add structured logging (pino/winston)
Health check	Basic, no DB test	Add DB connectivity check
Security headers	None	Add helmet middleware
Compression	None	Add compression middleware
API versioning	None	Add /v1/ prefix or header
Request tracing	None	Add request ID middleware
Error handling	Basic	Improve with stack traces in dev
Environment validation	None	Validate required env vars at startup
OpenAPI/Swagger	None	Add API documentation UI
Docker production	Only MySQL	Add full docker-compose.prod.yml

Recommended Fix Order

Phase 1: Critical for MVP (Before Feb 3)

1. Add Register toggle to LoginPage - Users can't sign up
2. Add Cancel button to RequestsPage - Incomplete workflow

3. Add `POST /inventory/return` - Missing core event type
4. Add `PUT /DELETE /vehicles` - Basic CRUD incomplete
5. Add pagination to `PartsPage` - Will break at scale

Phase 2: Core Improvements (Week 1 post-MVP)

6. Add `PUT/DELETE /locations`
7. Add `DELETE /interchange-groups`
8. Add `POST /requests/:id/reject`
9. Add vehicle detail page showing fitted parts
10. Add low stock alert threshold to parts

Phase 3: Production Hardening (Week 2)

11. Add structured logging
12. Add security headers (helmet)
13. Add compression
14. Add environment validation
15. Improve health check with DB ping
16. Add request ID tracing
17. Add error boundaries to React

Phase 4: Feature Parity with Competitors

18. Part images upload
19. Supplier management
20. Location-to-location transfers
21. CSV/PDF export
22. Dashboard analytics
23. Purchase order workflow

Quick Wins (Low Effort, High Impact)

1. **Cancel button on requests** - 10 min fix, completes workflow
 2. **Register on login page** - 20 min fix, critical for new users
 3. **Parts pagination** - 30 min fix, prevents scaling issues
 4. **Security headers** - 5 min fix (`npm i helmet` + 1 line)
 5. **Compression** - 5 min fix (`npm i compression` + 1 line)
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Files That Need Changes

Backend (ordered by priority)

- `src/routes/vehicles.routes.ts` - Add PUT/DELETE
- `src/routes/inventory.routes.ts` - Add return endpoint
- `src/routes/requests.routes.ts` - Add reject endpoint
- `src/routes/interchange.routes.ts` - Add group DELETE
- `src/controllers/vehicles.controller.ts` - Add update/delete handlers
- `src/controllers/inventory.controller.ts` - Add return handler
- `src/services/vehicles.service.ts` - Add update/delete logic
- `src/services/inventory.service.ts` - Add return logic
- `src/index.ts` - Add helmet, compression, better health check

Frontend (ordered by priority)

- `src/pages/LoginPage.tsx` - Add register form toggle
- `src/pages/RequestsPage.tsx` - Add cancel button
- `src/pages/PartsPage.tsx` - Add pagination
- `src/pages/VehiclesPage.tsx` - Show fitted parts
- `src/pages/InventoryPage.tsx` - Add search functionality
- `src/api/client.ts` - Add new API methods

Test Coverage Status

Test Type	Location	Status
Unit tests	<code>tests/unit/</code>	Exists but needs review
Integration tests	<code>tests/integration/</code>	Exists but needs review
E2E tests	None	MISSING
Frontend tests	None	MISSING

Database Schema Observations

The Prisma schema is well-designed with: -  Proper foreign keys and cascading deletes - 
Unique constraints where needed (SKU, vehicle combo) -  Indexes on frequently queried fields -

⚠️ `suppliers` table mentioned in CLAUDE.md but not in schema - ⚠️ `restock_rules` table mentioned but not in schema - ⚠️ No `low_stock_threshold` field on parts

Next Steps

Please review this analysis and let me know which items you want to prioritize. I recommend:

1. **If shipping MVP Feb 3:** Focus on Phase 1 only
2. **If you have more time:** Let's discuss Phase 2-4 priorities
3. **If you want me to start now:** I can begin with the Quick Wins

What would you like me to tackle first?